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Structure–function relationships of deep eutectic solvents for lignin extraction and chemical transformation†

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Lignin is the most abundant natural aromatic feedstock, and the conversion of lignin to value-added chemicals has drawn immense attention in biorefineries. Deep eutectic solvents (DESs) have been used for lignocellulosic biomass fractionation and lignin extraction due to their simple procedure, selective solubility of lignin, low cost, and high recyclability. The nature and number of functional groups in DESs remarkably influence the delignification and structural changes of lignin during treatment. Although many studies have investigated lignin extraction and the following chemical transformations during DES treatment, no review has illuminated the structure–function relationships between DESs and lignin. This review presents a systematic overview of important studies to provide insights into lignin extraction and chemical transformations by examining the relationship between the type and number of functional groups in DES constituents during pretreatment. Furthermore, various challenges and opportunities in the development of more sustainable and efficient lignin extraction are provided according to the remaining problems in this field.

1 Introduction

Due to the gradual depletion of fossil fuels and emerging environmental concerns, increasing attention has been paid to the development of alternative energy sources. Lignocellulosic biomass—mainly carbohydrates (cellulose and hemicelluloses) and lignin—is one of the most promising renewable feedstocks for producing chemicals and materials.[1–3] Over the past several decades, carbohydrate fractions have been successfully converted into many valuable products, such as bioethanol, levulinic acid, and fermentable sugars, which have achieved commercial profitability.[4–7] Biorefineries have widely utilized carbohydrate fractions, but significant amounts of lignin fractions remain underutilized.[8,9] Thus, large-scale and high-value applications of lignin are crucial for expanding biomass utilization and decreasing waste generation.

Lignin is a complex, three-dimensional amorphous biopolymer that is an abundant renewable aromatic resource.[10] Its chemical structure is composed of phenylpropane units, derived from three monolignols, namely sinapyl alcohol (S), coniferyl alcohol (G), and *p*-coumaryl alcohol (H), which are linked by various types of ether bonds (C–O bonds) and carbon–carbon bonds.[11] The use of lignin as a feedstock to prepare bio-based materials or to transform it into other chemicals has numerous advantages—it is the only natural aromatic feedstock and has high thermal stability and renewability.[12] However, lignin upgrading has been hampered due to biomass recalcitrance, lignin structural heterogeneity, inefficient extraction of lignin from biomass, and undesired lignin condensation during biomass pretreatment.[13,14] Therefore, an efficient fractionation process should enable the high-yield extraction and obtaining high-purity lignin, which is a key step towards realizing lignin valorization.

Traditional solvents for lignin extraction include dilute acids, alkali solvents, and organic solvents, which can solubilize lignin fragments (*e.g.*, γ-valerolactone, tetrahydrofuran, and methanol). However, these solvents normally suffer from equipment corrosion, volatility, high toxicity, or the need for high temperatures and pressures, which may decrease commercial profitability.[15,16] Recently, ionic liquids (ILs) have

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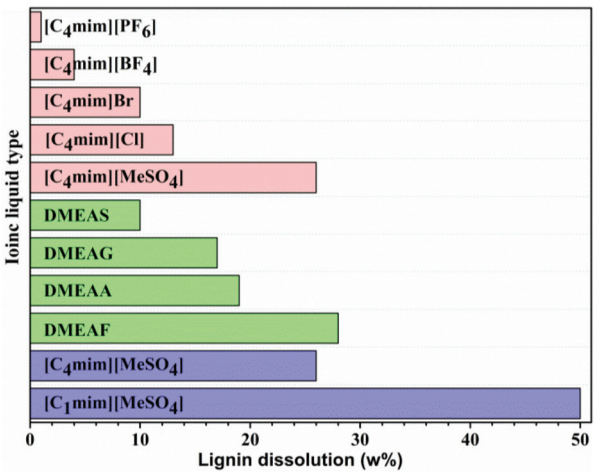


Fig. 1 Lignin dissolution in different ILs.[20] Note: [C_1mim][Cl]: 1,3-dimethylimidazolium chloride; [C_4mim][Cl]: 1-butyl-3-methylimidazolium chloride; [C_4mim][Br]: 1-butyl-3-methylimidazolium bromine; [C_4mim][BF_4]: 1-butyl-3-methylimidazolium tetrafluoroborate; [C_4mim][PF_6]: 1-butyl-3-methylimidazolium hexafluorophosphate; [C_4mim][$MeSO_4$]: 1-butyl-3-methylimidazolium methylsulfonate; DMEAF: *N*,*N*-dimethylethanolammoniumformate; DMEAA: *N*,*N*-dimethylethanolammoniumacetate; DMEAG: *N*,*N*-dimethylethanolammoniumglucuronidate; DMEAS: *N*,*N*-dimethylethanolammoniumsulfidate.

received extensive attention due to their ability to extract lignin in high yields.[17] The dissolution ability of lignin in different ILs is likely to be affected by the structure of ILs (Fig. 1). Despite being promising for lignin fractionation from lignocellulose, some disadvantages still restrict the practical applications of ILs in biorefinery processes, such as their high costs, complex synthesis procedures, and poor recoverability.[18,19] To overcome these limitations while retaining their desired features, deep eutectic solvents (DESs) have been investigated as novel green designer solvents.

DESs were first introduced by Abbott and co-workers and are eutectic mixtures that contain a hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA).[21] Francisco *et al.* found that DESs can selectively solubilize large amounts of lignin and negligible amounts of cellulose, suggesting that they could be suitable for extracting lignin from lignocellulose.[22] DESs have shown promise for lignocellulosic biomass processing, including biomass pretreatment, carbohydrate product conversion, and lignin extraction and upgrading.[23–27] Yu *et al.* first applied microwave-assisted DES pretreatment to realize the ultrafast fractionation of components from biomass in 3 min.[28,29] Several previous works have focused on lignocellulose fractionation using various DES systems.[30–33] Readers can consult the review articles on the chemical processes intensification of lignocellulose fractionation and lignin extraction and upgrading using various DES systems.[34–38] However, there is no comprehensive understanding of the correlation between the lignin extraction efficiency and the structure and properties of DESs during lignocellulose pretreatment. Lignocellulose delignification and lignin structure transformation strongly depend on the properties and number of HBD/HBA functional groups in DESs. Elucidating the chemical transformation of lignin in DESs is needed to design effective DESs to extract lignin with tailored properties.

This review provides a comprehensive summary of related studies that have explored the relationships between the performance of DESs and lignin extraction from three aspects: (1) functional group type and number of HBDs, (2) various categories of HBAs, and (3) DESs incorporated with catalysts or additives. Moreover, the correlation between the Kamlet–Taft parameters of DESs and the efficiency of lignin extraction is reviewed. Finally, the structural transformations of lignin under different DES pretreatment conditions are systematically discussed, and viewpoints on the opportunities and challenges for green and efficient lignin extraction according to existing problems are also given. Lignin extraction and chemical transformations in DESs are shown in Fig. 2.

2 Deep eutectic solvents (DESs)

DESs are eutectic mixtures generally consisting of a hydrogen bond donor (HBD) and a hydrogen bond acceptor (HBA), which combine *via* hydrogen bond (H-bond) interactions.[39,40] H-bonds within the mixtures are responsible for charge delocalization; thus, the melting point of DESs is much lower than that of the individual components.[41] For example, the mixture of ChCl (m.p. 302 °C) with urea (m.p. 132 °C) can lead to DESs with a melting point of 12 °C.[42] DESs possess many merits, including the availability of raw materials, easy synthesis, tunable physicochemical properties, and good recyclability.[43] DESs have been synthesized and used in different applications, such as catalysis, pharmaceutical applications, and biomass processing.[44]

Generally, DESs are classified into four main categories based on the nature of the complexing agent (Fig. 3). Type I DESs are formed from metal halides (*e.g.*, $AlCl_3$, AgCl, and $CuCl_3$) and quaternary ammonium salts (*e.g.*, choline chloride (ChCl), and have not been significantly applied to lignocellulose processing owing to the high melting points of non-hydrated metal halides.[39] Type II DESs mainly include hydrated metal halides (*e.g.*, $AlCl_3{\cdot}6H_2O$) and ChCl, which are more viable for industrial processes because most of these chemicals are low cost and are inherently insensitive to air/moisture. Type III DESs mainly consist of ChCl and various HBDs, such as carboxylic acids, alcohols, and amides. Type IV DESs include inorganic transition metals and HBDs (*e.g.*, urea), even though metal salts do not normally ionize in non-aqueous media.[45] Among the four types of DESs, Type III is the most common (especially in biomass processing), due to their low-cost starting materials, easy preparation, interactivity with water, low viscosity, and high biodegradability.[37,46]

The solvent properties greatly influence the efficiency of biomass pretreatment, delignification capability, and structural change of lignin. Thus, understanding the physico-

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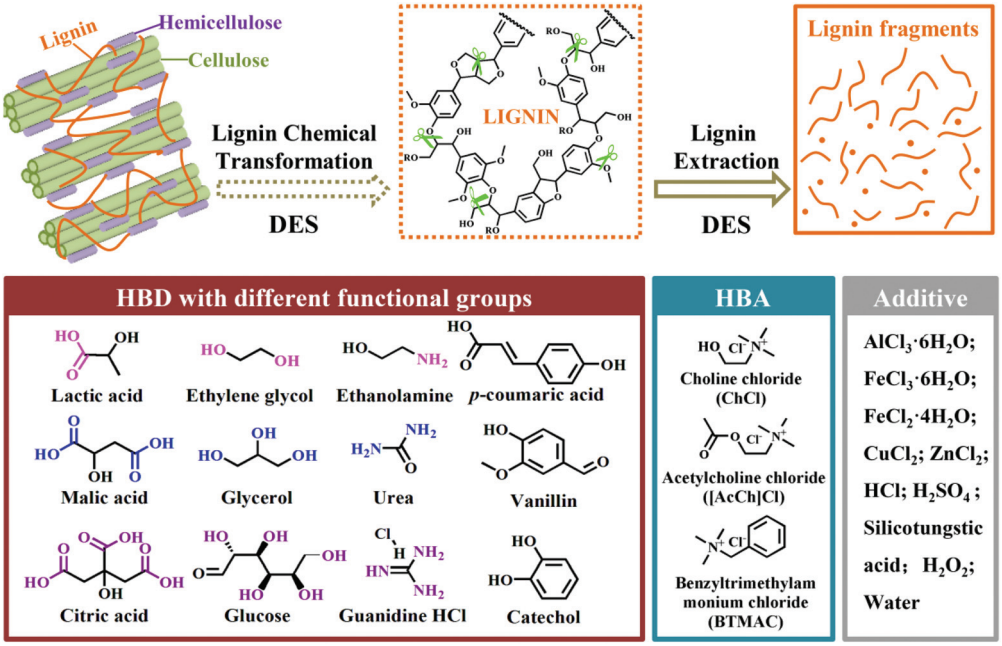


Fig. 2 Lignin transformation and extraction in different DES systems.

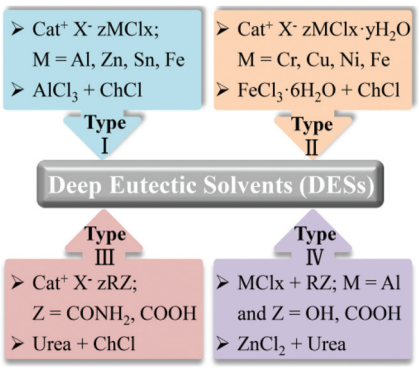


Fig. 3 General formula for the classification of DESs.[39] Note: Cat^+ is an ammonium, phosphonium, or sulfonium cation; X represents a Lewis base, generally a halide anion; Y is a Lewis or Brønsted acid; and z represents the number of y molecules that interact with the anion.

chemical properties of DESs is essential to fractionate lignin from lignocellulose. The important properties of DESs include the melting point, viscosity, conductivity, density, polarity, and surface tension. One of the important features of DESs is that they possess a lower melting point than their individual constituents, which is responsible for the charge delocalization due to the formation of H-bonds between HBAs and HBDs.[39] Hence, most DESs with low melting points exist as liquids at room temperature, allowing them to be used as solvents in biomass processing. In general, the melting point of DESs decreases upon increasing the H-bonding strength between mixtures.[47] However, the presence of extensive H-bonding networks within a mixture may increase the viscosity of DESs. Numerous studies have revealed that DESs with high viscosities are unfavorable for lignin extraction due to the low rates of heat and mass transfer during DES pretreatment.[43]

The high viscosity of DESs still poses a problem. Conductivity and surface tension are closely related to viscosity.[43,48] It has been reported that reducing the viscosity of DESs can promote its conductivity.[49] Moreover, the conductivity of DESs is inversely proportional to temperature.[50] In most cases, DESs have higher surface tensions than many conventional ILs.[51] The presence of water and low temperature can enhance the surface tension of DESs.[52] The surface tension of DESs strongly depends on the intermolecular forces between mixtures; thus, highly viscous DESs are fluids with high surface tensions.[53] Density also is an important property of DESs, as it can reveal the activity and molecular mobility of the solution. This information is helpful in some applications like lignin extraction processes. In most cases, DESs have a higher density than water.[37,53] It is generally observed that increasing the temperature or adding water to DESs lowers their density.[53–55] Overall, the understanding of the preparation, classification, and physiochemical properties of DESs facilitate their utilization in lignocellulose pretreatment and lignin extraction.

3 The effect of the structure–function relationships of DESs on lignin extraction

The most important step during lignin valorization is the efficient extraction/dissolution of lignin with favorable properties. DESs are highly effective toward lignin solubilization

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as compared to both cellulose and hemicellulose.[22,56,57] One of the reasons for this is that the DESs possess strong H-bonding networks, which contribute to the high interaction energy between DESs and lignins; however, the strong cohesive energy of cellulose may restrict cellulose dissolution in DESs. Accordingly, the DESs are able to act either as hydrotropes or cosolvents to improve the lignin solubilization.[58] The DES performance in lignin extraction is affected by many factors, such as pretreatment time, extraction temperature, DES type, solid-to-liquid ratio, and HBA/HBD mole ratio. Among these factors, the quality and quantity of functional group in DES components have the greatest influence. The DESs are usually prepared by mixing natural and renewable constituents, which determines the properties of the synthetic DESs and their behavior during lignin fractionation from biomass.

3.1 Functional group type and number in HBDs

Many HBDs including carboxylic acids, polyols, and amides are suitable for DES synthesis. The functional group of HBDs can interact with lignin, which affects the lignin extraction efficiency and chemical structure. Investigating the relationships between the types of functional groups in HBDs and lignin extraction is important to produce DESs to extract lignin with tailored properties. In addition, depending on the number of functional groups (*e.g.*, carboxyl groups) in HBDs, the lignin extraction yield has been reported to vary (Fig. 4).

3.1.1 Carboxyl-containing HBDs. Numerous HBDs with carboxyl groups exhibit better performance in biomass pretreatment and delignification than those containing other groups (*e.g.*, hydroxyl groups), regardless of the HBA used. Carboxyl-containing HBDs are more diverse, including formic acid, oxalic acid, and lactic acid, and most can remove >90% lignin from nearly all types of lignocellulose (poplar, Douglass fir, grass, and willow) when the molar ratio of HBD to HBA is high.[23,28,63,64] This excellent delignification capacity mainly arises due to the presence of more active protons supplied by carboxylic acids, which promotes the proton-catalyzed cleavage of chemical linkages in lignocellulose (*e.g.*, ether linkages in lignin and ester linkages in lignin-polysaccharide compounds). To further understand the influence of various carboxyl-containing HBDs on the performance of DESs, it is necessary to explore the structure–function relationships of various carboxyl-containing DESs.

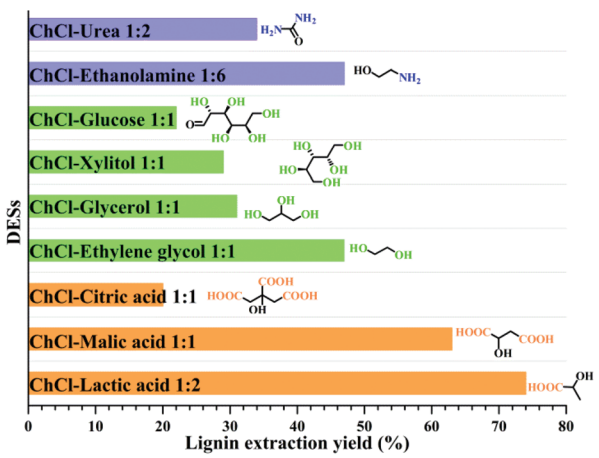


Fig. 4 Influence of HBD chemical structure on lignin extraction yield.[30,59–62]

The properties of carboxylic acid-based HBDs can affect the performance of DESs during lignin extraction. Previous research has shown that the lignin extraction ability improves with the acid strength of HBD. The acid dissociation constant (pK_a) of monocarboxylic acid–based DESs increases in the order: ChCl/formic acid, ChCl/lactic acid, ChCl/acetic acid, ChCl/butanoic acid, and ChCl/propionic acid, while the lignin extraction performance generally decreases.[59] However, ChCl/lactic acid (pK_a = 3.86) produced a greater pretreatment effect than a stronger acid, ChCl/malonic (pK_{a_1} = 2.83, pK_{a_2} = 5.69).[63]However, ChCl/malonic acid exhibited a better delignification performance from rice straw than ChCl/oxalic acid (pK_{a_1} = 1.25, pK_{a_2} = 4.14).[60] These results implied that the lignin fractionation efficiency was not necessarily related to the amount or strength of the acid when the HBD contained more than one carboxyl group.

The number of carboxyl groups in an HBD is positively correlated with the extent of delignification during DES pretreatment. ChCl/lactic acid (1 : 9 molar ratio) (a monocarboxylic acid) is capable of selectively dissolving 95% of lignin from poplar and achieving higher lignin extraction yield than ChCl/succinic acid (a dicarboxylic acid) and ChCl/citric acid (a tricarboxylic acid).[65] The result indicated that the monocarboxylic acid HBD possessed the optimal extraction capacity compared with the di- and tricarboxylic acids. This finding agrees well with Tan *et al.*, who showed that the lignin extraction efficiency declined as follows: ChCl/lactic acid > ChCl/malic acid > ChCl/citric acid at the same molar ratio.[59] One possible explanation for the weak delignification of di- and tricarboxylic acids is that the carboxyl groups in the HBD may easily be activated and thus release carbon dioxide at relatively high temperatures.[63] The study also found that extra carboxyl groups in HBDs formed more H-bonds and extensive dimer chains, which reduced the mass transfer rate of the solvent system.[66] This implied that efficient interactions between DESs and lignins were impaired when there was more than one carboxyl group in the HBD. Hence, the number of carboxyl groups in HBDs is inversely proportional to the delignification ability of the DESs.

3.1.2 Hydroxyl-containing HBDs. HBDs with hydroxyl groups are also widely used to synthesize DESs, including polyalcohol-based HBDs (*e.g.*, ethylene glycol, propylene glycol, and glycerol) and carbohydrate-based HBDs (*e.g.*, glucose, fructose, and xylitol).[67,68] This type of DES usually exhibits good biocompatibility with enzymes and microorganisms at near-neutral pH when mixed with a nontoxic HBA (*e.g.*, betaine, ChCl). Due to the neutrality of the hydroxyl-containing HBDs, most works have shown that polyalcohols or carbohydrate-based DESs have low efficiencies during lignin dissolution and fractionation, even under severe pretreatment conditions.

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Hydroxyl-containing HBD pretreatment is suitable for grass, which has a weaker bond energy of the covalent bond.[23,69,70] Hydroxyl groups in HBDs play a positive role during lignin extraction. Compared with the performance of DESs with the same number of carboxyl groups, ChCl/lactic acid ($C_3H_6O_3$) with a 33.5% lignin yield outperformed ChCl/propionic acid ($C_3H_6O_2$) with 20.4% and ChCl/malic acid ($C_4H_6O_5$) achieved a higher lignin yield (22%) than ChCl/succinic acid ($C_4H_6O_4$) (6%).[59] Therefore, the presence of hydroxyl groups in HBDs improves the performance of DESs during lignin extraction, due to interactions between the hydroxyl groups in the HBD and the free and etherified hydroxyl groups in lignin. Therefore, exploring how the hydroxyl groups in HBDs affect the performance of DESs will help improve pretreatment by DESs.

The number of hydroxyl groups in HBDs is inversely correlated with lignin removal. ChCl/glycerol showed worse lignin extractability than ChCl/ethylene glycol.[63,65] In another work, the pretreatment efficiency and lignin removability of polyalcohol-based DESs decreased in the order of ethylene glycol > glycerol > xylitol for polyol sequences when using lactic acid as the HBA.[60] Therefore, during DES pretreatment, a lower delignification rate was associated with a higher number of hydroxyl groups in the HBD. It was deduced that the extra hydroxyl groups in HBDs formed an extensive H-bond network between HBAs and HBDs, thus weakening the ability of the HBA to compete with intra-molecular bonding in the biomass components.[71] With the same number of hydroxyl groups in different positions, the delignification ability was also slightly different. For example, ChCl/1,3-propanediol separated more lignin from rice straw than ChCl/1,2-propanediol.[60] The author showed that the greater hydrophobicity of 1,3-propanediol-based DESs promoted lignin extraction using the hydrophobic phenylpropanoid skeleton.

3.1.3 Amine/amide-containing HBDs. To further understand the effect of functional groups in HBDs on the performance of DESs, HBDs with amine/amide groups, such as urea, imidazole, and ethanolamine, were explored for their lignin fractionation effects. For instance, corncob pretreated by ChCl/imidazole at 150 °C removed 88% of lignin.[72] Amines or amides present in HBDs improved lignin extraction by DESs, due to the loosened linkages between lignin and carbohydrates. A common feature of amine/amide-based DESs is their basicity, which is conducive for the extraction of lignin from lignocellulose. As the basicity decreased from ChCl/monoethanolamine, ChCl/diethanolamine, ChCl/*N*-methyldiethanolamine, to ChCl/urea, the lignin removal rate significantly decreased from 81.0%, 73.5%, 44.6%, to 27.7%, respectively.[73] The main reason for this trend was that the phenolic hydroxyl groups of lignin were deprotonated in alkaline solutions. In addition, the effect of the number of amine groups in the HBD on the lignin removal ability was also investigated to improve pretreatment efficiency. Lactic acid/formamide achieved a higher lignin removal rate from rice straw than lactic acid/urea,[60] which implied that the pretreatment performance of amine/amide-based HBDs showed an identical trend to hydroxyl-containing HBDs. HBDs with extra amines/amides or hydroxyls can form more intermolecular H-bonds with HBAs, which reduces the number of free and active groups capable of interacting with lignocellulose and lowers the lignin production efficiency during DES pretreatment.

3.1.4 Alkyl chain length of HBD. Few investigations have been conducted on the relationship between the aliphatic chain length and lignin extractability, although initial research suggests that the alkyl group position in the HBD appears to affect the lignin extraction efficiency. Methyldiethanolamine (44.6%) containing an alkyl group (methyl) showed a weaker lignin fractionation ability than diethanolamine (73.5%).[73] The main cause was the likely steric hindrance from the methyl group in methyldiethanolamine, which hampered the lignin fractionation efficiency. Moreover, the alkyl chain length of the HBD also influences lignin extraction. For monocarboxylic HBDs, the extracted lignin yield decreased as follows: ChCl/formic acid (62%) > ChCl/acetic acid (27%) > ChCl/propionic acid (20%) > ChCl/butanoic acid (14%) (Fig. 5).[59] For dicarboxylic HBDs, the lignin extraction rate decreased as follows: ChCl/oxalic acid (98.5%) > ChCl/malonic acid (56.5%) > ChCl/glutaric acid (34.3%).[63] Overall, HBDs containing shorter alkyl chains were more effective at delignification than those with longer chains, possibly due to electron donation from the alkyl group to the oxygen of the –OH group in the acid, which intensified the H-bonding strength.[74] Weaker acids ionize to a lesser extent; therefore, decreasing the chain length of the HBD will increase the ability of the solvent to donate protons, which improves the DES pretreatment performance. However, another research study has found that ChCl/lactic acid ($C_3H_6O_3$) produced a higher delignification efficiency than ChCl/glycolic acid ($C_2H_4O_3$) when pretreated under the same conditions.[60] It is likely that intramolecular H-bonds between adjacent functional groups in glycolic acid formed more easily than those in lactic acid, which made it difficult to cleave the various bonds in lignin. More efforts should be made to further understand the relationship between the alkyl chain length in HBDs and the lignin removal rate.

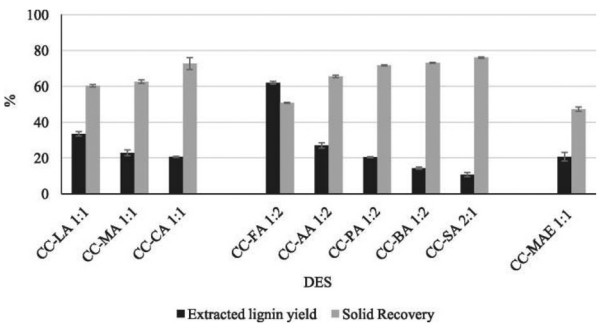


Fig. 5 Extracted lignin yield in different DESs.[59] Note: CC; choline chloride; LA: lactic acid ($C_3H_6O_3$); MA: malic acid ($C_4H_6O_5$); CA: citric acid ($C_6H_8O_7$); FA: formic acid (CH_2O_2); AA: acetic acid ($C_2H_4O_2$); PA: propanoic acid ($C_3H_6O_2$); BA: butanoic acid ($C_4H_8O_2$); SA: succinic acid ($C_4H_6O_2$); and MAE: maleic acid ($C_4H_4O_4$).

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3.1.5 Lignin-derived HBDs. Numerous lignin-derived phenolic compounds, such as resorcinol, vanillin, and catechol, have been evaluated as HBDs for synthesizing DESs in biomass pretreatment. Kim *et al.* tested ten lignin-derived phenolic compounds as HBDs to prepare novel DESs, and only four phenolic-based HBDs (4-hydroxybenzyl alcohol, catechol, vanillin (VAN), and *p*-coumaric acid (PCA)) formed a homogeneous solvent with ChCl.[75] The authors revealed that HBDs with methoxyl groups at the *ortho* position (G unit and S unit) synthesized almost no DESs compared with those without methoxyl groups due to the steric hindrance of the methoxyl groups adjacent to the hydroxyl groups. Among the four DESs, ChCl/PCA at 160 °C for 3 h achieved the highest lignin removal rate (60.8%), followed by ChCl/vanillin (52.5%), ChCl/catechol (49.0%), and ChCl/4-hydroxybenzyl alcohol (0.4%). With the introduction of phenolic-based HBDs by Kim and co-workers, research for lignin-derived compounds as DESs has received growing interest. For example, 64.1% lignin removal was achieved using ChCl/PCA to pretreat herb residues at 180 °C for 7 h.[76] This high delignification efficiency was attributed to the presence of more active protons donated by carboxyl groups in PCA, which facilitated the proton-induced bond cleavage in lignin. The phenyl group in HBDs allows the formation of more hydrogen bonds, which increases the stability of solvents and delignification.

To further develop the potential of lignin-derived DESs in biorefineries, Ragauskas groups utilized DES from lignin-derived phenolic aldehydes, namely ChCl/VAN and ChCl/4-hydroxybenzaldehyde, to pretreat transgenic biomass with aldehyde-rich lignin.[77] The author proved that abundant fermentable sugars were released to a great extent after lignin-derived DES pretreatment while the recovered lignin from DES can be further processed to supply more phenolic aldehydes for synthesizing DESs. This research proved that the integration of lignin-derived DESs with lignocellulose can achieve a closed-loop biorefinery, which is a sustainable bioeconomy (Fig. 6). In addition, 64.1% lignin removal was achieved using ChCl/PCA to pretreat herb residues at 180 °C for 7 h. To explore the effect of phenyl groups in HBDs on lignin fractionation, three types of phenolic-based HBDs (*e.g.*, phenol, α-naphthol, and resorcinol) were used to prepare DESs using ChCl as the HBA.[78] Resorcinol dissolved 48.2% of lignin, followed by phenol (40.4%) and α-naphthol (23.9%), which suggested that lignin solubility declined when two phenyl groups were present. Since a variety of lignin-derived compounds are available, further screening of such compounds will provide insight towards more desirable HBDs for lignin extraction and upgrading.

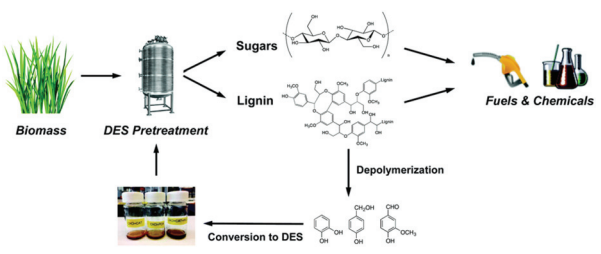


Fig. 6 A closed-loop biorefinery using lignin-derived DESs for biomass pretreatment.[75]

Thus far, the presence of the carboxyl group in HBDs could generally improve the delignification performance of DES compared to the hydroxyl or amine/amide group. When considering only the number of functional groups, the monocarboxylic acid HBDs possess a better lignin extraction capacity than the di- and tricarboxylic acid HBDs. The lignin removal performance of hydroxyl- and amine/amide-containing HBDs showed an identical trend with carboxyl-containing HBDs. Besides, HBDs containing shorter alkyl chains are more effective at delignification than those with longer chains. The functional group type and number in HBDs obviously affects the lignin extraction ability of DESs, indicating the importance of selecting suitable HBDs to form DESs with excellent delignification capability. An in-depth understanding of the relationship between the functional groups in HBDs and the lignin extraction efficiency will guide future DES designs and will largely determine the final delignification efficiency.

3.2 Categories of HBAs

HBAs also play important roles in the pretreatment performance of DESs. The lignin extraction performance of ChCl/formic acid was compared with that of pure formic acid, and the results showed that ChCl/formic acid removed 40.7% of lignin, while pure formic acid only removed 9.0% from herbal residues.[79] Hence, the DESs display better lignin fractionation than the corresponding acids alone. The high efficiency of DES delignification is attributed to the synergistic effects of HBDs and HBAs.[80] Currently, HBAs mainly include a variety of ammonium, phosphonium, and sulfonium salts.[39] One of the most widespread HBAs for synthesizing DESs is ChCl, due to its biodegradability, low cost, and nontoxicity. ChCl can also rapidly form DESs and is effective in lignin extraction. Previous research has found that the highly electronegative Cl^- in ChCl tended to form H-bonds with the hydroxyl groups in lignin, which improved lignin removal.[81] Other research studies have shown that the halide counterion in the HBA facilitates the breakage of β-*O*-4′ bonds and also prevents lignin self-condensation.[82,83] However, when containing the same halogen ions, betaine-based DESs only dissolved 12% lignin at the same substrate loading, while ChCl-based DESs dissolved 100% of lignin.[84] This implies that the efficiency of lignin removal is closely influenced by the coordinating ability of cations in the HBA. To further explore the influence of the HBA structure on the performance of the DESs, the biomass solubilities of three HBAs (ChCl), acetylcholine chloride ([AcCh]Cl), and benzyldimethyl(2-hydroxyethyl)ammonium chloride ([BzCh]Cl), were investigated using rice straw.[68] The lignin solubility decreased in the order of ChCl > [AcCh]Cl > [BzCh]Cl regardless of the HBD used. In another work, four HBAs with different functional groups, ChCl, benzyltrimethylammonium chloride (BTMAC), benzyltriethylammonium

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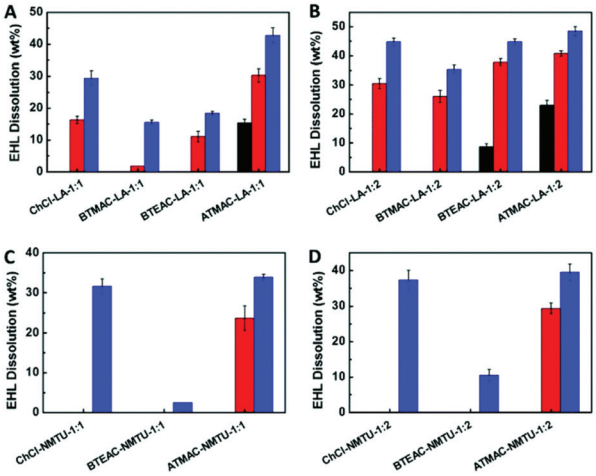


Fig. 7 The solubility of lignin in DESs with different HBDs.[85] Note: *EHL* represents enzymatic hydrolysis lignin.

chloride (BTEAC), and allyltrimethylammonium chloride (ATMAC), were used to prepare DESs to dissolve lignin.[85] The results suggested that the lignin solubility order was ATMAC > ChCl > BTMAC/BTEAC, regardless of the HBD used (Fig. 7). This indicated that the functional groups enhanced lignin dissolution in the order: C=C > OH > benzene ring, possibly because the C=C in ATMAC contributed to the formation of more H-bonds than OH in ChCl and benzene ring in BTMAC, which possibly enhanced solvent interactions with lignin through H-bonds. Meanwhile, the length of the alkyl groups in HBAs also plays a profound role in the delignification efficiency. Guo *et al.* found that BTMAC/LA had a relatively high lignin removal rate compared with BTEAC/LA.[86] This is because BTMAC has a shorter alkyl chain and, therefore, exhibited less steric hindrance during lignin removal compared with BTEAC. Furthermore, some works have shown that certain HBDs can be used as HBAs. When using lactic acid as HBA, the biomass dissolution efficiency of the DESs was better than that using ChCl-based DESs, regardless of the HBD employed.[60] This shows that a suitable selection of HBA and HBD largely determines the DES pretreatment severity and biomass fractionation efficiency. Hence, identifying the optimal combinations of DESs is the first step for their efficient utilization.

3.3 Intensified DESs with catalysts/additives

Most DESs are effective during biomass pretreatment but require rigorous pretreatment conditions, such as high temperatures and long treatment times. To improve the DES pretreatment performance, extra components are added to DESs as catalysts or additives, such as acids, organic solvents, and water (Table 1). For example, the lignin extraction capacity of polyol-based DESs (*i.e.*, ChCl/glycerol, ChCl/ethylene glycol) was significantly improved by adding a small amount of H_2SO_4.[69,87] The high performance of acidified DESs was ascribed to the higher availability and reactivity of acidic protons in the DES environment. In another study, mixtures of ChCl/glycerol and different Lewis acids were used to remove lignin from grass, and the use of $AlCl_3$, $FeCl_3$, and $CuCl_2$ dramatically improved delignification due to the enhanced H-bond acceptor strength and acidity in acidified DESs.[88] Particularly, by incorporating $AlCl_3{\cdot}6H_2O$ into ChCl/glycerol, the lignin extraction efficiency was markedly increased from 3.61% to 95.46%.[71] This increase in delignification was because $AlCl_3{\cdot}6H_2O$ is an anion donor and an active acidic site holder, which provides active protons and acidic sites that facilitate proton-catalyzed bond cleavage in lignin (Fig. 8). In addition to inorganic acids and Lewis acids, other acids have

Table 1 Comparison of the lignin removal rate between DESs and DESs with catalysts/additives.[88–90,92,98]

	Catalysts/additives					
DESs	Types	Content	Feedstocks	Temperature (°C)	Time (h)	Lignin extraction yield (%)
ChCl : oxalic acid (1 : 2)	—	—	Onion root	110	6	39.5
	$AlCl_3{\cdot}6H_2O$	0.2 M	Onion root	110	6	75.5
	—	—	Kraft lignin	120	4	2.7
	Water	5 wt%	Kraft lignin	120	4	9.4
ChCl : urea (1 : 2)	—	—	Onion root	110	6	49.5
	$AlCl_3{\cdot}6H_2O$	0.2 M	Onion root	110	6	76.6
	—	—	Kraft lignin	120	24	9.4
	Water	5 wt%	Kraft lignin	120	24	10.6
ChCl : glycerol (1 : 2)	—	—	*Pennisetum*	120	6	4.0
	$FeCl_3{\cdot}6H_2O$	0.2 M	*Pennisetum*	120	6	90.0
	$FeCl_2{\cdot}4H_2O$	0.2 M	*Pennisetum*	120	6	39.0
	$ZnCl_2$	0.2 M	*Pennisetum*	120	6	41.0
	$CuCl_2$	0.2 M	*Pennisetum*	120	6	88.0
	$AlCl_3{\cdot}6H_2O$	0.2 M	*Pennisetum*	120	6	91.0
	PTSA	3 wt%	Switchgrass	120	0.1	30.1
	Silicotungstic acid	Trace	*Miscanthus × giganteus*	120	3	89.5

Note: "1 : 2" represents the molar ratio of HBA to HBD. "0.2 M" represents the molar ratio of catalysts/additives and HBA. "Trace" represents less than 1 wt% of silicotungstic acid.

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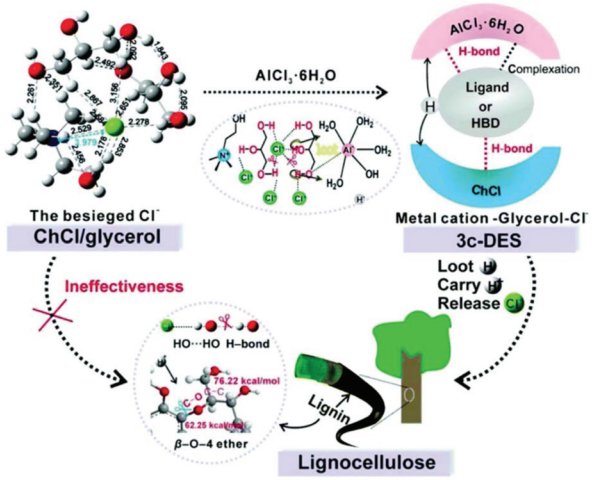


Fig. 8 ChCl/glycerol with $AlCl_3{\cdot}6H_2O$ enhanced lignin extraction from biomass.[71]

been explored to improve DES pretreatment performance. For example, *p*-toluenesulfonic acid (PTSA), a strong solid organic acid, is a strong proton donor, and ChCl/ethylene glycol with PTSA removed 82% of lignin from switchgrass in only 6 min at 120 °C.[89] Silicotungstic acid, an eco-friendly heteropoly acid, is a strong Brønsted acid with good stability and catalytic activity. The use of neutral DESs with silicotungstic acid at 120 °C for 3 h achieved 89.5% lignin removal and 97.3% enzymatic digestibility of the remaining carbohydrate pulp.[90] These results demonstrate that there is a strong synergism between DESs and various acids during pretreatment, and acidified DESs exhibit better performance than the non-acidified ones.

Water appears to increase the effectiveness of DES pretreatment. When the molar ratio of HBA to HBD was the same, the addition of water could improve the solubility capacity of lignin, regardless of DES types.[91] Kumar *et al.* studied the influence of water addition to DESs on lignin extraction and found that water addition (5 wt%) to ChCl/lactic acid slightly improved lignin extraction from rice straw.[84] Another study also found that lignin removal was enhanced due to the addition of water to DESs,[92] which significantly decreased their viscosity, which was responsible for improving the solubilization capacity of DESs.[93,94] Water molecules are also involved in the hydrolysis of esters or ethers.[95] However, there has been no clear study on how water interacts with DESs to increase the pretreatment effectiveness.

An increasing number of chemical compounds have been incorporated into DESs to boost their performance. Lignin is highly soluble in nonpolar solvents such as butanol, and these solvents are favorable for lignin extraction. A delignification of 23–31% was obtained with ChCl/oxalic acid, while 50% delignification was achieved with ChCl/oxalic acid using *n*-butanol at 120 °C for 1 h.[96] These results show a strong synergism between organic solvents and ChCl/oxalic acid during lignin removal. Similarly, a combination of alkaline hydrogen peroxide (AHP) and ChCl/urea resulted in 18.99% delignification from oil palm fronds (OPFs), which was 1.6-fold higher than delignification with pure DES.[97] The synergistic effect between AHP and ChCl/urea was confirmed by characterizing pretreated OPFs. Further screening of suitable compounds may be required to prepare desirable DESs for lignocellulosic biomass fractionation or lignin extraction. A better understanding of the synergy between DES components will provide guidance for fine-tuning solvent properties for lignin extraction.

4 Kamlet–Taft solvatochromic parameters of DESs

Although a variety of DESs has been synthesized, there are no systematic studies on the relationship between DES properties and industrial applications. The chemical properties of DESs can be described by the Kamlet–Taft (K–T) solvatochromic parameters, which serve as a quantitative tool.[99] The K–T parameters are defined by three solvent parameters, including H-bond-donating ability (α), H-bond-accepting ability (β), and polarizability (π^*). These parameters can be used to explain the internal connections between DES properties and their ability to fractionate lignin from lignocellulose.[23] For example, Yu *et al.* studied the correlations between the K–T parameters of DESs and the lignin extraction rate and found that the β value was positively correlated with the delignification rate.[71] The authors explained that a high H-bond basicity improved the ability of the DES to break the H-bond network in lignin. In another work, ChCl/oxalic acid with a high H-bond acidity (α = 1.31) broke the structure of the lignin–hemicellulose complex, while ChCl/urea and ChCl/glycerol did not.[28] This is because the acidic protons dissociated from DESs catalyzed the cleavage of ethers and esters in lignocellulose. Moreover, the lignin-dissolving ability was related to the difference between the values of α and β (α–β), which shows a net H-bond-donating ability of DESs. For example, the lignin solubility/extraction capability of DESs increased at higher α–β values.[85,100] These results imply that DESs with high α–β values could form strong H-bonds with lignin, which was favorable for the degradation of lignin. In addition to α and β values, the π^* value of DESs also affects the lignin extraction ability. Studies have shown a close correlation between lignin extraction/solubility and π^* values;[23,101] however, there is still a lack of a systematic overview for specific interactions between lignin extraction and the K–T parameters of DESs.

To understand which of the three parameters (α, β, and π^*) is most important for delignification, the correlation between the K–T parameters and lignin extraction rate are summarized based on the existing data (Fig. 9). It was observed that the relationship between the lignin extraction ability and the K–T parameters varied depending on DES composition. An inductive analysis shows that the α and β values of DESs are more strongly related to the lignin extraction ability than the π^*

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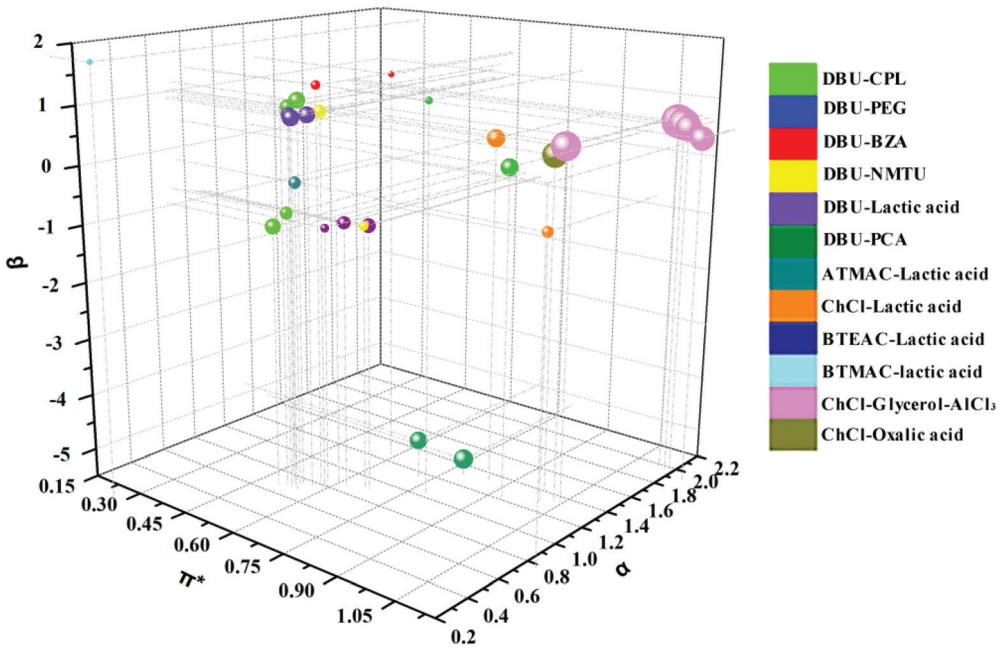


Fig. 9 The correlation between lignin extraction and the K–T solvatochromic parameters of DESs (the ball size represents the lignin extraction yield). Note: DBU: 1,5-diazabicyclo[5.4.0]-5-undecene, CPL: ε-caprolactam, PEG: polyethylene glycol 200, BZA: benzamide, NMTU: *N*-methylthiourea, PCA: 4-pyridine carboxamide, ATMAC: allyltrimethylammonium chloride, BTMAC: benzyltrimethylammonium chloride, and BTEAC: benzyltriethylammonium chloride.

value; therefore, it is plausible that higher α and β values of DESs indicate a higher lignin removal efficiency.

5 Chemical transformation of lignin during DES pretreatment

Due to the good performance of DESs in achieving high enzymatic saccharification yields and delignification from lignocellulose, the mechanisms about the interaction of DESs with lignin have recently gained more attention. For ChCl-based DESs, chloride anions are capable of forming strong H-bonds with the hydroxyl groups of lignin fragments, which break the intramolecular H-bond network of lignin and promote lignin fractionation.[71] C–H···π and O···H bonding interactions contributed to the high interaction energy between ChCl/lactic acid and lignin, thus enhancing lignin dissolution in DESs.[102] An insight into the structural transformations of lignin during DES pretreatment is not only essential to understand and overcome biomass recalcitrance but it is also pivotal to design optimal pathways for lignin valorization. During DES processing, both depolymerization and repolymerization can occur; thus, the final structure of the DES-extracted lignin may be the net effect of the two reactions whose outcome strongly depends on the processing method, which includes: (1) acidic DES pretreatment, (2) alkaline DES pretreatment, and (3) neutral DES pretreatment. This following section summarizes the possible chemical reactions during DES pretreatment, with a focus on the fate of β-*O*-4′ linkages in the lignin substructure.

5.1 Acidic DES pretreatment

Acidic DESs exhibit excellent performance during lignin extraction; thus, investigation into the structural transformations of lignin in acidic DESs is necessary for lignin valorization. The possible reactions of the lignin substructure during acidic DES pretreatment are illustrated in Fig. 10. The main reaction is lignin depolymerization, especially the breakage of β-*O*-4′ bonds. Alvarez-Vasco *et al.* have reported that when treated with ChCl/lactic acid, guaiacylglycerol-β-guaiacyl ether (a β-*O*-4′ bond lignin dimeric model compound) got converted into guaiacol and Hibbert's ketone.[23] This result demonstrated that the breakage of β-*O*-4′ bonds favored lignin depolymerization, which consequently facilitated lignin fractionation from lignocellulose. Similar research also proved that acidic DESs can selectively cleave ethers and degrade a small portion of C–C bonds (*e.g.*, β-5′ and β-β′) in lignin, which facilitates lignin separation from lignocellulose.[65,103] The prevailing depolymerization pathway during acidic DESs is the destruction of the β-*O*-4′ linkages among phenylpropane units *via* mild acid-catalyzed bond cleavage.[23,34,100,104] The most important step is the protonation of the α-OH group to form a stable benzylic carbocation. Then, the benzylic carbocation loses the terminal γ-CH_2OH group to generate α–β unsaturated compounds and formaldehyde, which further promotes the cleavage of ether bonds (Route I). In an alternate pathway, an enol ether intermediate is formed after an elimination reaction between the

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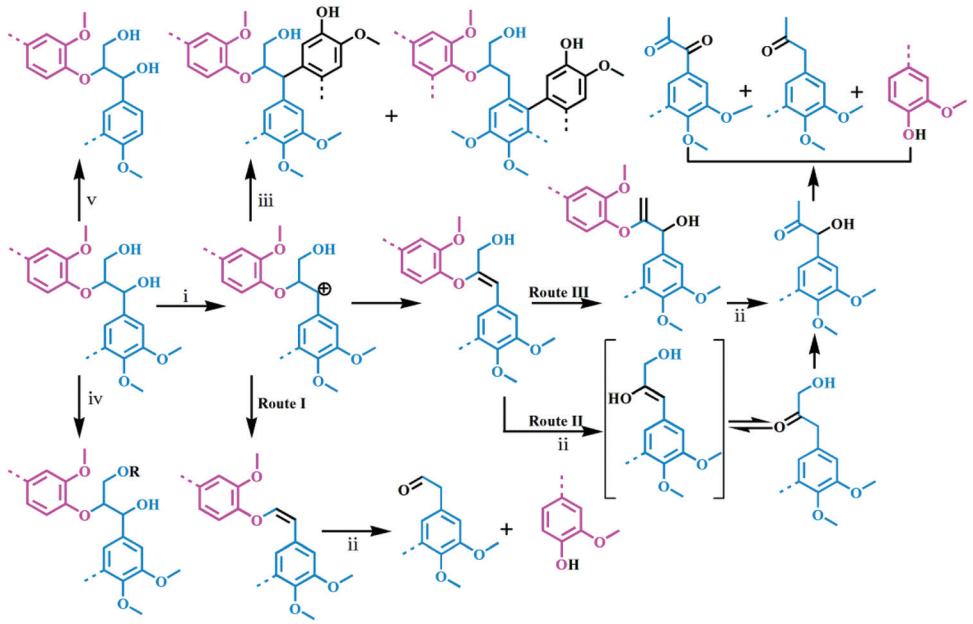


Fig. 10 Proposed lignin reaction pathways during acidic DESs pretreatment:[23,100,104,107,108,110] (i) dehydration; (ii) depolymerization; (iii) repolymerization; (iv) acylation; and (v) demethoxylation.

α-carbocation and β-H. The subsequent hydrolysis of the acid-labile enol ethers causes the breakage of the β-*O*-4′ bonds, followed by the modification of the Hibbert's ketone moiety in DES systems, finally producing depolymerized compounds, such as monoketone and diketone compounds (Route II). Another pathway is the allylic rearrangement of an enol ether intermediate followed by hydrolysis, which leads to the cleavage of β-*O*-4′ bonds and produces monoketone and diketone compounds (Route III). During acidic DES pretreatment, without an interacting anion, it proceeds through Route I with the elimination of formaldehyde; when the DES anion (halide ion) interacts with the γ-OH group, Route II and III are favored, similar to dilute acids and acidic ILs.[105–107] Certain acidic DESs oxidize the α-OH group, followed by the acylation of the γ-OH group, which favors ether bond cleavage to yield diketones (not depicted in Fig. 10).[100,108] It is worth noting that lignin depolymerized products in DES systems contain diketone compounds, which have not been found in other acid pretreatments up to now.

In addition to depolymerization, repolymerization of lignin fractions can also occur during acidic DES pretreatment. For example, Shen and co-workers found that the chemical shift of the guaiacyl unit of lignin changed after ChCl/lactic acid processing, which suggested the formation of new C–C bonds *via* acid-catalyzed recondensation.[109,110] Repolymerization commonly occurs through the formation of covalent bonds between the α-C of the benzylic carbocation and electron-rich positions on aromatic rings to obtain diphenylmethane compounds. However, the coupling between the electrophilic positions on the aromatic rings of benzylic carbocations and electron-rich positions of aromatic rings forms biphenyl compounds. Acidic DESs not only result in the condensation of lignin fraction (*e.g.* ChCl/lactic acid) but also preserve lignin from condensation and produce non-condensed lignin (*e.g.*, ChCl/boric acid). Non-condensed lignin was reported by Hiltunen's groups, with softwood sawdust as the raw material and ChCl/boric acid as the pretreatment solvent at 95 °C.[111] The reason for the production of non-condensation lignin is that boric acid, a Lewis acid, can form esters with side-chain hydroxyl groups in lignin, thus reducing oligomerization and repolymerization reactions.

In certain acidic DES systems, esterification occurs between carboxyl groups in acidic DESs and γ-OH in lignin substructures. Several research groups have reported that γ-acetylated signals in DES-extracted lignin were observed by 2D-HSQC NMR (Fig. S1†), which indicated the acylation of lactic acid during ChCl/lactic acid pretreatment.[65,82,100,109] In addition to the above reactions, demethoxylation and dehydration also frequently occur during DES pretreatment.[100,110] However, the exact mechanism is complex and not clearly understood.

5.2 Alkaline DES pretreatment

Alkaline solvents are often used to extract lignin from biomass because lignin is an alkaline-soluble biopolymer.[107] For example, chemical fractions of holocellulose, α-cellulose, and acid-insoluble lignin were pretreated with ChCl/urea, and lignin exhibited a higher selective solubility compared with other chemical fractions.[112] Alkaline DES pretreatment destroys lignin–carbohydrate compounds, disrupts lignin by breaking the ether bonds, and eventually results in lignin depolymerization/repolymerization.[97] Similarly, pretreatment with basic ILs resulted in the cleavage of ether bonds, along with demethoxylation, dehydration, and repolymerization primarily through β-β′and β-5′ linkages.[113] In another study, basic

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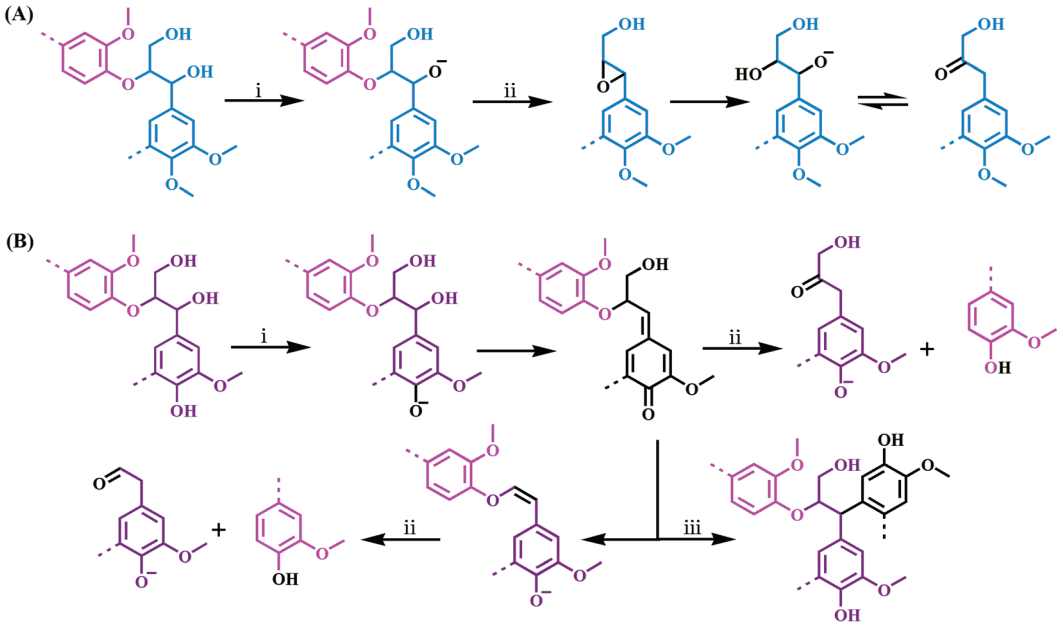


Fig. 11 Proposed reaction pathways of lignin during alkaline DES pretreatment in (A) phenolic units and (B) non-phenolic units:[97,107] (i) dehydration; (ii) depolymerization; and (iii) repolymerization.

IL (cholinium lysinate ([Ch][Lys])) pretreatment resulted in a similar reaction pathway in which the phenolic β-ether moiety formed a quinone methide intermediate that then underwent depolymerization and/or repolymerization.[114] When using traditional basic solvents (*e.g.*, NaOH), the reactivity of the β-*O*-4′ bonds within the lignin substructure highly depends on the subunits type, and a difference has been observed between phenolic subunits (free phenolic OH-groups) and non-phenolic subunits (etherified phenolic OH-groups).[115] Therefore, the transformation mechanism of lignin during alkaline DES pretreatment can be postulated to proceed as follows (Fig. 11): for non-phenolic subunits, the breakage of β-*O*-4′ bonds under alkaline conditions may slowly form an epoxide intermediate (A); and phenolic subunits in the presence of alkaline DESs may form a quinone methide–type intermediate due to the deprotonation of phenolic OH-groups, which then undergoes depolymerization and repolymerization (B).

5.3 Neutral DES pretreatment

In contrast to acidic and alkaline DES pretreatment, neutral DES pretreatment has a minor effect on lignin extraction and structural transformation. Yu *et al.* revealed that the intramolecular H-bonds in ChCl/glycerol were mutually constrained by anionic H-bonds ($[Cl(glycerol)]^-$) and cationic H-bonds ($[Ch(glycerol)]^+$), in which ChCl/glycerol had weakly competing interactions towards the linkages in the network of the lignin–carbohydrate complex.[71] Wan and co-workers showed that lignin recovered from ChCl/glycerol pretreatment partially preserved the β-*O*-4′ linkages and had a similar structure to protolignin.[69] In another study, the lignin extracted after ChCl/ethylene glycol pretreatment exhibited a low molecular weight and preserved partial β-*O*-4′ bonds;[87,116] however, few studies have examined the lignin transformation mechanism using alkaline DESs. After near-neutral IL pretreatment, lignin possessed the highest β-*O*-4′ content, the largest molecular weight, and the minimum degree of condensation compared to that after acidic/basic IL pretreatment;[114] therefore, pretreatment with neutral DESs may only slightly change the lignin structure. However, a comprehensive study of the structural transformation of lignin in neutral DES systems is required.

6 Conclusions and prospects

The applications of DESs for lignin extraction and conversion have been widely recognized. Although a variety of DESs has been widely studied and used in biomass pretreatment, there is still a need for a rigorous investigation on the structure–function relationship of DESs during processing to increase the pretreatment efficiency. The properties of ideal DESs for lignin extraction are still unclear. Research on DES synthesis and its practical applications are still in their infancy, and thus more in-depth studies are required to increase DES pretreatment efficiency in future works. To achieve the ultimate goal of efficient lignin extraction and upgrading in terms of depolymerization and functionalization, there is still much work to do, as follows:

(1) DESs are ideal solvents for lignin fractionation and catalytic upgrading. Nevertheless, most DES screening methods are based on time-consuming trial and error. The directed design of DESs based on their intended purpose is promising. Besides, if lignin-derived compounds can be further used to synthesize DESs, a promising approach may be opened up to utilize lignin.

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(2) The K–T solvatochromic parameters are a quantitative tool that can express the chemical properties of DESs. These parameters may be used to predict the pretreatment performance of DESs. Systematic studies concerning how the K–T solvatochromic parameters of DESs affect lignin extraction are important, but so far only a few studies have investigated this topic.

(3) Although various DESs have been widely applied in lignocellulose pretreatment, previous work has revealed that DES selectivity and lignin extraction yield vary greatly in different DES systems. Investigating the properties of DESs is vital to ensure their feasibility for lignin extraction and elucidating the interactions between DESs and lignin will greatly improve lignin extraction and valorization.

(4) Traditional DES pretreatment processes carry out lignocellulose fractionation and conversion separately, which significantly increases the operating costs, the number of process operational steps, and waste discharge. Hence, it is necessary to develop a novel DES pretreatment technology that integrates lignin fractionation, enzymatic saccharification, and fermentation in a single vessel. Moreover, effective DES recycling methods should be investigated.

Conflicts of interest

There is no conflict to declare.

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